INCH-POUND

A-A-50727A
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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION
GENERATOR, SIGNAL
(2 GHz to 26 GHz)

The General Services Administration has authorized the use of the Commercial Item Description (CID) for all federal agencies.

1. SCOPE. This Commercial Item Description (CID), describes a laboratory standard quality bench piece of electronic test equipment which is a bench-mount with rack-mount option, solid-state, general purpose, 2 GHz to 26 GHz, signal generator. This CID is meant as a minimum requirement for the signal generator, in which only those manufacturers that meet or surpass the following requirements are to be supplied per this CID.

2. SALIENT CHARACTERISTICS

2.1 Equipment covered by this CID shall be commercially available equipment and may modified to the extent necessary to meet the following description. The equipment shall be Class 3, in accordance with MIL-PRF-28800. This CID makes provision for bid sample testing in accordance with the latest MIL-PRF-28800. The equipment shall be capable of operation within the accuracies, limits, and specifications herein.

2.2 Safety. The equipment shall be Underwriters Laboratories (UL) listed and approved and/or shall comply with the safety requirements of MIL-PRF-28800 for the classification stated herein.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: WR-ALC/LEEC, 295 Byron Street, Robins AFB, GA 31098-1611

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2.3 Electrical Power Sources. The equipment shall operate from nominal commercial, military, and shipboard single-phase power sources of 100, 110, 120, 220, and 240 volts ac (± 10%) at line frequencies of 48 Hz to 66 Hz.

2.4 Power Consumption. The maximum power consumption of the equipment shall not exceed 450 VA.

2.5 Dimensions and Weight. The size and weight of the equipment shall be consistent with current commercial capabilities.

2.6 Signal Input and Output Connectors. The RF output connector type shall be 2.92-mm female, and all other signal input and output connectors shall be type BNC female.

2.7 Performance characteristics

2.7.1 Digital Input/Output. The Measuring Receiver and Synthesizer/Local Oscillator shall incorporate an IEEE 488.1 standard I/O capability. The Interface Function subset implemented shall have or exceed the following capabilities:

INTERFACE FUNCTION	MINIMUM ACCEPTABLE SUBSET
Source Handshake	SH1
Acceptor Handshake	AH1
Talker	T1 or TE1
Listener	L1 or LE1
Remote Local	RL1
Parallel Poll	PP0
Device Clear	DC1
Device Trigger	DT1
Service Request	SR1

The signal generator shall use bi-directional data transmission in ASCII coded format.

2.7.2 Frequency. The minimum frequency range of the signal generator shall be 2 GHz to 26 GHz.

2.7.2.1 Resolution. The frequency resolution of the signal generator shall be at least 4 KHz throughout the entire frequency range.

2.7.2.2 Accuracy. The frequency accuracy shall be based on the aging rate of the internal reference oscillator. The aging rate shall be no more than 1 X 10^{-9} per day after 10 days of continuous operation.

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2.7.3 Harmonics. Harmonics of the signal generator shall be no more than –25 dBc over the entire frequency range.

2.7.3.1 Sub-harmonics. Sub-harmonics of the signal generator shall be no more than –25 dBc to 18 GHz and no more than –20 dBc from 18 GHz to 26 GHz.

2.7.3.2 Spurious non-harmonics. The spurious non-harmonics of the signal generator shall be at least –55 dBc over the entire frequency range.

2.7.4 Absolute Single-Side Band (SSB) Phase Noise. The absolute SSB phase noise of the signal generator at 18 GHz with 100 KHz offset shall be no more than -90 dBc per Hz.

2.7.5 Output. The output level range of the signal generator into 50 O nominal shall be +3 dBm to –95dBm.

2.7.5.1 Output Resolution. The output resolution of the signal generator shall be at least 1 dB.

2.7.6. Amplitude Modulation (AM). The AM sensitivity of the signal generator shall be 1 Volt (V) peak-to-peak (P-P) for essentially 100% modulation.

2.7.7. AM Rate. The AM rate of the signal generator shall be at least 20 Hz to 10 KHz (3 dB bandwidth).

2.7.8 Frequency Modulation Deviation. The deviation of the signal generator shall be 20 MHz with a 10 Hz to 100 KHz FM input, 10 MHz with a 100 KHz to 1 MHz FM input.

2.7.9 Pulse Modulation. The on/off ratio of the signal generator for pulse modulation shall be 30 dB or greater.

2.7.9.1 Rise and Fall Time. The rise and fall time of the signal generator shall be between 10% and 90% of the pulse shall be 25 nanoseconds or less.

2.7.9.2 Repetition Rate. The minimum repetition rate of the signal generator for a carrier frequency of 2 GHz to 26 GHz shall be 10 Hz to 50 KHz.

2.7.9.3 Pulse Width. The minimum pulse width of the signal generator shall be at least 100 nano-seconds.

2.7.10 Sweep Modes. The signal generator shall have start/stop or delta F (span) sweep functions, and the following sweep modes: manual, automatic, and single sweep.

2.7.11 Ramp Output. The signal generator shall have a 1 V to 10 V ramp output proportional to frequency from start to stop.

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2.7.12 Rack-mount Capability. The signal generator shall have rack-mount capability available as an option for mounting in a standard 19 inch Electronic Industries Alliance (EIA) rack.

2.7.13 Manuals. The signal generator shall be delivered with operator, maintenance, and illustrated parts manuals. Format and quantity shall be as specified in the contract or order.

3. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. PRODUCT CONFORMANCE

4.1 Product Conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards and quality assurance practices, and be the same product offered for sale in the commercial marketplace. Small businesses that cannot offer products with established commercial market acceptability may compete by submission of bid samples. The government reserves the right to require proof of such conformance.

4.2 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the specified tolerances using conversion tables contained in the latest revision of ASTM-SI10, and all other requirements of this CID are met.

4.3 Exceeded Metric Products. If a product manufactured to metric dimensions exceeds the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

5. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Sources of documents.

6.1.1 Government Documents. Military Specifications, Standards, and Handbooks referenced herein may be obtained from the Standardization Documents Order Desk, 700 Robbins Ave, Bldg 4, Section D, Philadelphia, PA 19111-5094.

6.1.2 Underwriters Laboratories (UL).Copies of UL standards can be obtained from UL Corporate Headquarters, 333 Pfingsten Road, Northbrook IL 60062-2096. Phone: 1-847-272-8800. Fax 1-847-407-1395.

6.1.3 IEEE standards. A copy of IEE standards can be obtained from IEEE Operations Center, 445 Hoes Lane, PO Box 1331, Piscataway, NJ 08855-1331. Phone: 1-800-678-4333 (Toll-Free, USA & Canada), 1-732-981-0060 (Worldwide), Fax: 1-732-981-9667.

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6.1.4 EIA standards. Copies of EIA standards can be obtained from EIA, COMM 2000, 1414 Brook Drive, Downers Grove, IL 60515, and U.S.A. Phone 1-800-UL33503 or 1-888-853-3503. Fax 1-888-8533-3512.

6.1.5 Code of Federal Regulations (CRF). Copies of the CFR may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC, 20402.

6.1.6 ASTM standards. Copies of ASTM standards can be obtained from ASTM, 100 Bar Harbor Drive, West Conshohocken, PA 19428-2959. Phone 1-610-832-9500. Fax 1-610-832-9555.

MILITARY INTEREST

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